

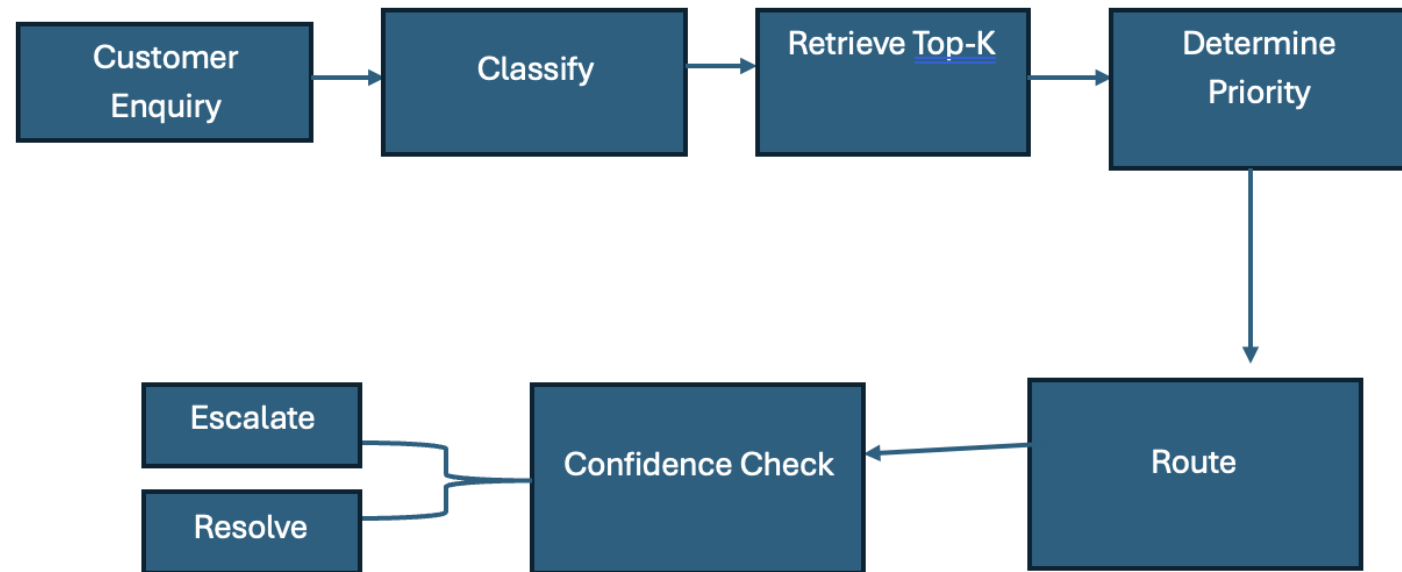
Smart Inquiry Triage Assistant Srikar Kodi

BWM Case Study 2026

Business Problem

- Every customer inquiry currently needs a human to read it, categorize it, decide urgency, route
- Goal: automate the obvious cases, escalate the genuinely uncertain ones to a human, not replace judgment, augment it

Architecture



6-node LangGraph pipeline: one real Edge, two outcomes

Where LLM is used and where it isn't?

- Only one of five pipeline steps calls an LLM: writing the resolution note
- Category, priority, and routing are all embeddings + deterministic logic. No LLM guessing.
- Why: structured fields need to be reliable and repeatable; an LLM asked to classify can drift between two near-identical inputs
- The LLM's one job is exactly what LLMs are good at natural language, not structured decisions.

Confidence Score design

- $\text{Confidence} = 0.5 * \text{category_score} + 0.5 * \text{agreement}$
- category_score = How close the enquiry is to its predicted category's definition
- agreement = what fraction of the retrieved similar share the same category
- Two independent signal combined, either alone can be confidently wrong.

Making the graph a real graph

- Original version: one conditional edge, but both branches pointed to the same node
- Escalation logic was hiding inside an if — not really using what LangGraph is for
- Fixed: split into two separate nodes, escalate and resolve
- Verified independently and confirmed the escalate path never calls the LLM at all

Calibrating the confidence threshold

- Default threshold was 0.5, never tested, just a placeholder
- Ran a sweep (0.20 - 0.60) against the historical dataset, scoring accuracy vs. escalation rate
- Old default was escalating 43% of correctly classified inquiries and moved to 0.35
- Accuracy hit 100% at 0.35 and never improved further and everything above that just escalates more for nothing

Stress testing with multi categorical queries

- Built a separate test probing queries that deliberately mix 2–3 categories
- Infotainment + warranty + service query: classifier margin 0.002
- Confidence dropped to 0.18, correctly escalated, the system knew it wasn't sure
- Found a bonus signal: classifier margin tracks ambiguity well, not yet used, good next step

A limitation I found but did not hide

- One query buried a real safety issue inside two ordering-related clauses
- Confidence came out at 0.72 - did not escalate
- Root cause is structural: single-label classification can't split mixed intents.
- Documented in the README, not fixed, proposed solution: a safety-keyword pre-check before classification

Live Demo

- Clean query -> auto-resolves, shows the full pipeline output
- Ambiguous query -> escalates, walk through why in real time
- Off-topic query -> escalates at confidence 0.08
- Drag the threshold slider live to show the trade-off directly

RoadMap

- Hybrid classifier: blend the taxonomy-based score with a kNN vote from retrieval
- Add classifier margin as a third confidence signal
- Safety-keyword pre-check ahead of classification for multi-intent queries

Thank You!